

# Shreyas Punacha

Postdoctoral Scholar

University of Washington School of Dentistry

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## Research interests

Systems Neuroscience, Excitable Media, Black-hole Thermodynamics

## Education

- 2015–2021 **National Institute of Technology Karnataka**, Surathkal, India  
Ph.D. in Physics  
Thesis: *Excitation wave dynamics and their interaction with external fields*  
Mentor: Dr. T. K. Shajahan
- 2012–2014 **Mangalore University**, Mangalore, Karnataka, India  
M.Sc. in Physics
- 2009–2012 **Vivekananda College**, Puttur, Karnataka, India  
B.Sc. in Physics, Mathematics & Chemistry

## Honors and scholarships

- 2017–2021 Senior Research Fellowship, Ministry of Human Resource Development, India
- 2015–2017 Junior Research Fellowship, Ministry of Human Resource Development, India
- 2009–2012 Basic Science Fellowship, Department of Collegiate Education, Government of Karnataka, India

## Teaching experience

- 2015 – 2017 Teaching Assistant, B.Tech Physics Course at NITK Surathkal
- 2017 – 2019 Teaching Assistant, M.Sc Computational Physics Course at NITK Surathkal
- 2015 – 2020 Laboratory Teaching Assistant, M.Sc Physics Laboratory at NITK Surathkal
- 2015 – 2020 Teaching Assistant, B.Tec Physics Laboratory at NITK Surathkal

## Research experience - Postdoctoral Scholar

November 2022–  
Present

### **Neuroscience of Aging (2 publications)**

Mentor: Fritzie Arce-McShane, University of Washington, Seattle.

The project distinguishes changes in the orofacial sensorimotor-cognitive neuronal network during healthy aging from those in pathological aging, aiming to identify early indicators of Alzheimer's disease (AD) and other age-related dementias in alignment with the NIH mission to detect individuals at risk for late-onset AD. Using a multidisciplinary approach in biomechanics, neurophysiology, and computational modeling, we study sensation–movement interactions by recording neural activity from chronically implanted microelectrode arrays across multiple cerebral cortex regions in rhesus macaques and tracking tongue and mandible kinematics with high-resolution biplanar videoradiography. This work is conducted in collaboration with the Washington National Primate Research Center (WaNPRC) at the University of Washington.

June 2015–  
October 2021

### **Low-Voltage Cardiac Defibrillation (7 publications)**

Mentor: Shajahan T K, National Institute of Technology Karnataka, Surathkal.

During my doctoral research, I used mathematical modeling and nonlinear dynamics to control spiral waves that drive life-threatening cardiac arrhythmias using low-voltage electric fields. I combined experiments on pattern formation in the Belousov–Zhabotinsky reaction with nonlinear dynamics theory and computer simulations of spiral and scroll waves to develop low-voltage defibrillation strategies. I focused on ventricular fibrillation, a leading cause of sudden death in industrialized nations that claims about 300,000 lives annually in the United States. This work could transform current therapies by using low-amplitude electrical stimuli instead of high-energy defibrillatory shocks, and it also offers insights relevant to other excitable media in biophysical fields such as neuroscience and cardiology.

June 2019–  
Present

### **Black Hole Thermodynamics (10 publications)**

Mentor: Ajith K M, National Institute of Technology Karnataka, Surathkal.

My research in black hole physics focuses on their thermodynamic properties and phase transitions across various theoretical frameworks. I have studied photon orbits and their impact on phase transitions in regular AdS black holes, as well as the effects of nonlinear electrodynamics on the dynamics and kinetics of these transitions. My work also explores dynamic phase transitions in black holes within massive gravity and examines the role of Lyapunov exponents in the phase structure of Lifshitz and hyperscaling violating black holes. I have also investigated the thermodynamics and Joule-Thomson expansion of 4-D Gauss-Bonnet AdS black holes, integrating higher-order curvature corrections. These studies collectively enhance our understanding of black hole stability and phase behavior in diverse gravitational models.

## Talks

October 2025

Neural Encoding of Tongue Kinematics In Swallowing  
Oral Health Sciences Retreat, University of Washington, Seattle

October 2024

Mapping Orofacial Regions of the Brain Using Intracortical Microstimulation  
Oral Health Sciences Retreat, University of Washington, Seattle

- November 2023 Effects Of Aging On Lingual Movements During Swallows  
Society for Neuroscience (SfN), Washington DC
- October 2023 Effects Of Aging On Lingual Movements During Swallows  
Oral Health Sciences Retreat, University of Washington, Seattle
- September 2016 Pattern Formation In Nature By Reaction Diffusion Processes  
Department of Physics National Institute of Technology Karnataka Surathkal
- February 2017 Reaction Diffusion Processes  
Department of Physics, National Institute of Technology Karnataka Surathkal

## Poster Presentations

- Oct – 2018 Shreyas Punacha, Anupama Sebastian and T K Shajahan.  
Unpinning of Spiral Waves From Multiple Obstacles  
Conference on Nonlinear Systems and Dynamics (CNSD), Jawaharlal Nehru University, New Mehrauli Road, Delhi
- Jan – 2024 Shreyas Punacha and Fritzie Arce-McShane  
Effects Of Aging On Lingual Movements During Swallows  
Oral Health Sciences Research Day, University of Washington, Seattle
- March – 2024 Shreyas Punacha and Fritzie Arce-McShane  
Effects Of Aging On Lingual Movements During Swallows  
WaNPRC Scientific Symposium, Orin Smith Auditorium, University of Washington South Lake Union Campus, Seattle.
- April – 2024 Shreyas Punacha and Fritzie Arce-McShane  
Effects Of Aging On Lingual Movements During Swallows  
Biomedical Research Awareness Day (BRAD), Office of Animal Welfare, University of Washington, Seattle.
- May – 2024 Shreyas Punacha and Fritzie Arce-McShane  
Effects Of Aging On Lingual Movements During Swallows  
CoNectome, Computational Neuroscience Center, University of Washington, Seattle
- March – 2026 Edric Tsang, Shreyas Punacha and Fritzie I Arce-McShane  
Effects of Lingual Nerve Afference on Cortical Orofacial Motor Representation  
University of Washington, Seattle
- March – 2026 Zhiyuan Yang, Victoria Hosack, Shreyas Punacha, Derric Tang, Lilah Favour, Jing-Sheng Li, and Fritzie Arce-McShane  
*Neuromechanics of Temporomandibular Joint During Mastication*  
IADR/AADOCR/CADR General Session & Exhibition, San Diego, CA

## Publications

- [1] M. S. Ali, A. Naveena Kumara, K. Hegde, C. L. A. Rizwan, S. Punacha, and K. M. Ajith. **Grey-body factor for an electrically charged regular-de Sitter black holes in d-dimensions.** *Eur. Phys. J. C* **85.4** (2025), p. 394. DOI: [10.1140/epjc/s10052-025-14121-0](https://doi.org/10.1140/epjc/s10052-025-14121-0).
- [2] S. J. C., K. R., K. Hegde, K. M. Ajith, S. Punacha, and A. N. Kumara. **Perturbations of black holes surrounded by anisotropic matter field.** *Phys. Rev. D* **111.6** (2025), p. 064034. DOI: [10.1103/PhysRevD.111.064034](https://doi.org/10.1103/PhysRevD.111.064034).

- [3] K. Hegde, A. N. Kumara, C. L. A. Rizwan, M. S. Ali, K. M. Ajith, and S. Punacha. **Thermodynamics, photon sphere and thermodynamic geometry of Ayón-Beato–García space–time.** *Int. J. Mod. Phys. A* **40.27** (2025), p. 2550115. DOI: [10.1142/S0217751X25501155](https://doi.org/10.1142/S0217751X25501155).
- [4] S. Punacha, K. Huang, and F. I. Arce-McShane. **Effects of Healthy Aging on Tongue-Jaw Kinematics During Feeding.** *Journal of Oral Rehabilitation* (2025).
- [5] K. R., D. D., K. M. Ajith, K. Hegde, S. Punacha, and A. N. Kumara. **Euclidean thermodynamics and Lyapunov exponents of Einstein–Power–Yang–Mills AdS black holes.** *Eur. Phys. J. C* **85.11** (2025), p. 1364. DOI: [10.1140/epjc/s10052-025-15064-2](https://doi.org/10.1140/epjc/s10052-025-15064-2).
- [6] K. Hegde, A. Naveena Kumara, C. L. Ahmed Rizwan, A. K. M., M. S. Ali, and S. Punacha. **Thermodynamics, phase transition and Joule–Thomson expansion of 4-D Gauss–Bonnet AdS black hole.** *Int. J. Mod. Phys. A* **39.21** (2024), p. 2450080. DOI: [10.1142/S0217751X24500805](https://doi.org/10.1142/S0217751X24500805).
- [7] A. N. Kumara, S. Punacha, and M. S. Ali. **Lyapunov exponents and phase structure of Lifshitz and hyperscaling violating black holes.** *JCAP* **07** (2024), p. 061. DOI: [10.1088/1475-7516/2024/07/061](https://doi.org/10.1088/1475-7516/2024/07/061).
- [8] A. Sebastian, P. Sibeesh, S. Amrutha, S. Punacha, and T. Shajahan. **Effect of electric field chirality on the unpinning of chemical waves in the Belousov–Zhabotinsky reaction.** *Chaos, Solitons & Fractals* **186** (2024), p. 115209.
- [9] U. P. Suresh, K. R., K. M. Ajith, K. Hegde, S. Punacha, and A. N. Kumara. **Spinning LQG black hole as a particle accelerator** (Aug. 2024). arXiv: [2408.15828 \[gr-qc\]](https://arxiv.org/abs/2408.15828).
- [10] S. Amrutha, A. Sebastian, P. Sibeesh, S. Punacha, and T. Shajahan. **Theory and experiments of spiral unpinning in the Belousov–Zhabotinsky reaction using a circularly polarized electric field.** *Chaos: An Interdisciplinary Journal of Nonlinear Science* **33.6** (2023).
- [11] A. N. Kumara, S. Punacha, K. Hegde, C. L. A. Rizwan, K. M. Ajith, and M. S. Ali. **Dynamics and kinetics of phase transition for regular AdS black holes in general relativity coupled to nonlinear electrodynamics.** *Int. J. Mod. Phys. A* **38.29n30** (2023), p. 2350151. DOI: [10.1142/S0217751X23501518](https://doi.org/10.1142/S0217751X23501518).
- [12] T. K. Safir, A. N. Kumara, S. Punacha, C. L. Ahmed Rizwan, C. Fairroos, and D. Vaid. **Dynamic phase transition of black holes in massive gravity.** *Annals Phys.* **458** (2023), p. 169480. DOI: [10.1016/j.aop.2023.169480](https://doi.org/10.1016/j.aop.2023.169480).
- [13] S. Amrutha, A. Sebastian, P. Sibeesh, S. Punacha, and T. Shajahan. **Mechanism of spiral wave unpinning in the Belousov–Zhabotinsky reaction with a DC electric field.** *The Journal of Physical Chemistry C* **126.46** (2022), pp. 19618–19626.
- [14] A. Sebastian, S. Amrutha, S. Punacha, and T. Shajahan. “Dynamics of chemical excitation waves subjected to subthreshold electric field in a mathematical model of the Belousov-Zhabotinsky reaction”. In: *Nonlinear Dynamics and Applications: Proceedings of the ICNDA 2022*. Springer, 2022, pp. 1241–1249.
- [15] A. Naveena Kumara, C. L. Ahmed Rizwan, S. Punacha, K. M. Ajith, and M. S. Ali. **Photon orbits and thermodynamic phase transition of regular AdS black holes.** *Phys. Rev. D* **102.8** (2020), p. 084059. DOI: [10.1103/PhysRevD.102.084059](https://doi.org/10.1103/PhysRevD.102.084059).

- [16] S. Punacha, N. K. A, and T. Shajahan. **Theory of unpinning of spiral waves using circularly polarized electric fields in mathematical models of excitable media.** *Physical Review E* **102.3** (2020), p. 032411.
- [17] S. Punacha, M Abhishek, and T. Shajahan. **Unpinning spiral wave anchored to two obstacles.** *Indian Academy of Sciences* **2** (2019).
- [18] S. Punacha, S. Berg, A. Sebastian, V. I. Krinski, S. Luther, and T. Shajahan. **Spiral wave unpinning facilitated by wave emitting sites in cardiac monolayers.** *Proceedings of the Royal Society A* **475.2230** (2019), p. 20190420.

### Mentorship

- 2015–2017     **NITK Surathkal / Mentor**  
Master's Thesis: Unpinning of spiral waves pinned to heterogeneities in excitable media - M Abhishek
- 2016–2018     **NITK Surathkal / Mentor**  
Master's Thesis: Spiral wave unpinning facilitated by wave emitting sites in cardiac monolayers - Anupama Sebastian
- 2017–2019     **NITK Surathkal / Mentor**  
Master's Thesis: Detachment of spiral waves from heterogeneity using continuous electric field: a one dimensional approach - Aswin Ramesh
- 2017–2019     **NITK Surathkal / Mentor**  
Master's Thesis: Tip reflection of spiral waves at the boundaries - Greeshma G S
- 2018–2020     **NITK Surathkal / Mentor**  
Master's Thesis: Simulation of scroll wave dynamics in excitable media - Tejasri Gururaj
- 2025–2026     **University of Washington, Seattle / Mentor**  
Poster: Effects of Lingual Nerve Afference on Cortical Orofacial Motor Representation - Edric Tsang
- 2025–2026     **University of Washington, Seattle/ Mentor**  
Poster: Texture-Dependent Sensorimotor Patterns Under Scopolamine and Nerve Block - Hannah Ryu

### Peer review experience (25 manuscripts)

- [1]     *Physical Review E*
- [2]     *Physica Scripta*
- [3]     *Modelling and Simulation in Materials Science and Engineering*
- [4]     *Nonlinear Analysis: Real World Applications*
- [5]     *Physical Review Research*
- [6]     *International Journal of Electrochemical Science*

- [7] *Physical Review Letters*
- [8] *Fortschritte der Physik*
- [9] *Physics Letters B*
- [10] *ACS Omega*
- [11] *Physical Review D*
- [12] *Frontiers in Astronomy and Space Sciences*
- [13] *Frontiers in Behavioral Neuroscience*
- [14] *European Physical Journal C*

### Technical skills

**Programming:** MATLAB, Python, C, LaTeX, Git, Parallel Programming, HPC

**Tools:** DeepLabCut, 3D Slicer, Brainsight, Mayavi, Message Passing Interface

**Experimental Neuroscience:** Primate handling and training, Surgery planning, Neural recordings, X-ray Reconstruction of Moving Morphology (XROMM).

### Other interests

Coordinated a science outreach initiative “The Principia” for high school students in Karnataka and Kerala States of India.

Amateur bird photographer.

Ebirder - An active participant in [eBird](#), a citizen science project for bird observation from Cornell Lab of Ornithology.